

The Derivative Gain Gap: A Delayed-Feedback Control Model for Adolescent Idiopathic Scoliosis

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Abstract

Purpose: Despite decades of investigation, the etiology of adolescent idiopathic scoliosis (AIS) remains incompletely understood. Proprioceptive abnormalities and associations with peak height velocity have been reported but lack a unifying mechanistic framework. We propose a control-theoretic model in which growth-velocity-dependent degradation of the derivative (rate-sensing) gain in the trunk postural controller creates a transient instability window—the “derivative gain gap”—that may explain the clinical epidemiology of AIS.

Methods: The human trunk was modeled as an inverted pendulum governed by delayed proportional-derivative (PD) feedback, with controller gains scaled from published postural control parameters. Delay-differential equation simulations were performed across a parameter space of derivative gain (K_d), proprioceptive delay (τ), and growth velocity. Sex-specific growth curves from published CDC/WHO data were used to model growth-phase trajectories through the stability landscape.

Results: The relationship between derivative gain and critical delay (τ^*) was non-monotonic, with an optimal K_d maximizing the stability corridor. Growth-velocity-dependent K_d degradation drove the system trajectory across the Hopf bifurcation boundary exclusively when rapid growth velocity (>6 cm/yr) co-occurred with elevated proprioceptive processing delay ($\tau > 200$ ms). The model predicted female vulnerability at 10.6–11.8 years and male vulnerability at 12.2–14.2 years, consistent with published clinical data. Simulated external bracing, modeled as haptic augmentation of K_d , restored stability without altering intrinsic neuromuscular parameters.

Conclusions: This computational model generates the hypothesis that AIS onset reflects a transient derivative gain gap during peak growth velocity. The model reproduces sex-specific timing, female preponderance, and spontaneous stabilization at skeletal maturity, and suggests testable predictions for clinical validation.

1 Introduction

Adolescent idiopathic scoliosis (AIS) affects 2–4% of children aged 10–16 years, with a marked female preponderance in curves exceeding 30 degrees [1, 2]. Despite its clinical significance and decades of investigation, the etiology of AIS remains incompletely understood. Current etiological theories span biomechanical [4, 5], neurological [6, 8], genetic [10], and hormonal domains [11], yet no single framework satisfactorily accounts for the full clinical profile: adolescent onset, correlation with peak height velocity (PHV), pronounced female predominance, and spontaneous stabilization at skeletal maturity.

Several clinical observations suggest that postural control dysfunction may play a role in AIS pathogenesis. Proprioceptive abnormalities have been documented in AIS patients, including impaired trunk repositioning accuracy and altered vibration perception thresholds [6, 7]. Balance studies have demonstrated increased postural sway in AIS patients compared with age-matched controls [12]. Critically, AIS onset clusters around PHV—approximately 11.5 years in girls and 13.5 years in boys [13]—suggesting that rapid growth may interact with neuromuscular control to create a vulnerability window.

The human postural control system can be modeled as a feedback controller stabilizing an inherently unstable load (the upright trunk). Peterka’s seminal work established that quiet standing is well described by a delayed proportional-derivative (PD) controller with experimentally measurable gains and time delays [3]. The proportional gain (K_p) acts as a stiffness-like restoring term, while the derivative gain (K_d) provides rate-dependent damping analogous to viscous resistance. Importantly, the time delay (τ) inherent to neural signal transmission and processing imposes fundamental limits on the achievable stability margin [14, 15].

A growing body of control-theoretic literature has examined postural stability in the context of time-delayed feedback [16–18]. These studies demonstrate that increased feedback delay narrows the stability region in gain–delay parameter space, and that a Hopf bifurcation—the transition from stable equilibrium to sustained oscillation—occurs when delay-induced phase lag exceeds a critical threshold. However, to our knowledge, no prior work has specifically examined how *growth-velocity-dependent changes* in controller gains

interact with delay to create a transient instability window during adolescence.

In this study, we develop a computational model of trunk postural control in which the derivative gain (K_d) undergoes transient degradation during rapid growth, reflecting the hypothesis that rapid elongation of the proprioceptive pathway degrades rate-sensing performance before adaptive recalibration can occur. We demonstrate that this “derivative gain gap” creates a non-trivial stability landscape in which the system trajectory crosses the Hopf bifurcation boundary at ages consistent with clinical AIS onset. We further show that the model naturally reproduces sex-specific timing differences and suggests a mechanistic reinterpretation of brace therapy as haptic augmentation of the degraded derivative gain.

We emphasize that this work is hypothesis-generating. The model makes specific, falsifiable predictions that warrant clinical testing but has not been validated against patient data.

2 Methods

2.1 Mathematical Model

The trunk was modeled as an inverted pendulum with a single angular degree of freedom (θ), stabilized by a delayed PD controller representing the combined action of proprioceptive sensing, central neural processing, and muscular actuation. The equation of motion is:

$$I\ddot{\theta}(t) = mgL\theta(t) - b\dot{\theta}(t) - K_p\theta(t - \tau) - K_d\dot{\theta}(t - \tau) \quad (1)$$

where I is the moment of inertia of the trunk about the lumbar pivot ($0.8 \text{ kg}\cdot\text{m}^2$), m is effective trunk mass (25 kg), g is gravitational acceleration (9.81 m/s^2), L is the distance from the pivot to the center of mass (0.30 m at baseline), b is passive viscoelastic damping ($1.0 \text{ N}\cdot\text{m}\cdot\text{s/rad}$), K_p is the proportional (stiffness) gain, K_d is the derivative (damping) gain, and τ is the total sensorimotor delay.

The gravitational toppling torque is $mgL = 73.6 \text{ N}\cdot\text{m/rad}$, requiring the proportional

gain to satisfy $K_p > mgL$ for static stability. Default parameters were $K_p = 120 \text{ N}\cdot\text{m}/\text{rad}$ and $K_d = 8 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$, consistent with the ranges reported by Peterka [3] after scaling for the trunk (as opposed to full-body) configuration.

The sensorimotor delay τ was treated as a composite parameter encompassing afferent conduction, central processing, and efferent conduction times. Published values for postural control delays range from 100–300 ms [3, 20], with adolescents showing longer delays than adults [19]. During rapid growth, τ was modeled as increasing due to elongation of afferent pathways and developmental lag in myelination:

$$\tau(v_g) = \tau_0 + \alpha \cdot v_g \quad (2)$$

where $\tau_0 = 150 \text{ ms}$ is the baseline delay, v_g is the growth velocity in cm/yr , and $\alpha = 10 \text{ ms}/(\text{cm}/\text{yr})$ is a scaling coefficient representing the delay increment per unit growth velocity. The value of α was chosen to produce delay increases consistent with the reported range of 100–300 ms for postural control delays across development [3, 19]; it should be regarded as an assumed parameterization pending direct measurement in longitudinal adolescent cohorts.

2.2 Growth-Velocity-Dependent Derivative Gain Degradation

The central hypothesis of this model is that the derivative gain K_d undergoes transient degradation during periods of rapid growth. This is motivated by the observation that derivative (rate) sensing requires calibrated proprioceptive signal processing, which may be disrupted when the musculoskeletal geometry changes rapidly. The degradation function was modeled as:

$$K_d(t) = K_{d,\text{base}} \cdot \left[1 - \beta \cdot \frac{v_g(t)}{v_{g,\text{max}}} \right] \quad (3)$$

where $K_{d,\text{base}} = 8 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$ is the fully calibrated derivative gain, $v_g(t)$ is the age-dependent growth velocity, $v_{g,\text{max}}$ is the peak growth velocity for the given sex, and β is the maximal fractional degradation at PHV. We adopted $\beta = 0.6$ as a representative

value; no direct measurement of growth-dependent derivative gain degradation currently exists in the literature, and this parameter should be understood as a postulated quantity whose clinical value remains to be determined empirically. At PHV, this implies the effective K_d drops to 40% of its baseline value.

2.3 Hopf Bifurcation Analysis

The stability boundary of the linearized system (Eq. 1) was determined analytically by seeking the conditions under which the characteristic equation admits purely imaginary roots $s = j\omega$. Substituting into the characteristic equation:

$$Is^2 + bs - mgL + (K_p + K_d s)e^{-s\tau} = 0 \quad (4)$$

and separating real and imaginary parts yields two equations in two unknowns (ω, τ) for given K_p and K_d . The critical delay $\tau^*(K_d)$ at which the Hopf bifurcation occurs was computed by numerical root-finding over a frequency grid $\omega \in [0.5, 40]$ rad/s.

2.4 Simulation Protocol

Delay-differential equations were integrated using a fixed-step Euler method with $dt = 0.5$ ms and a history buffer of length $\lceil \tau/dt \rceil$. Each simulation ran for 15–20 s of simulated time from an initial perturbation $\theta_0 = 0.05$ rad ($\approx 2.9^\circ$). Stability was assessed by computing the amplitude ratio $R = \text{RMS}_{\text{late}}/\text{RMS}_{\text{early}}$, where the early and late windows comprised the first 15% and last 15% of the time series, respectively. A simulation was classified as unstable if $R > 2.5$ or $\max |\theta| > 1.0$ rad.

The following parameter sweeps were performed:

1. **Sweep A:** Fine-grained K_d sweep (50 values, 1–50 N·m·s/rad) at five K_p levels (90, 100, 120, 150, 200 N·m/rad), determining τ^* for each combination.
2. **Sweep B:** Two-dimensional stability map over a 50×76 grid of $K_d \in [1, 50]$ and $\tau \in [0, 0.300]$ s.

3. **Sweep C:** Growth-phase transients with time-varying K_p (Gaussian dip simulating the growth spurt) at eight K_d values.
4. **Sweep D:** Optimal K_d envelope across 45 K_p values from 80 to 300 N·m/rad.
5. **Sweep E:** Stochastic sensitivity analysis with additive Gaussian noise ($\sigma = 0, 1, 3, 5$ N·m) across seven K_d values (30 trials each).

2.5 Sex-Specific Growth Data

Growth velocity curves were derived from published CDC and WHO reference data [21, 22]. Female PHV was set at 11.5 years (mean velocity 8.3 cm/yr) and male PHV at 13.5 years (mean velocity 9.5 cm/yr), consistent with longitudinal growth studies [13, 23]. Growth velocity was modeled as a Gaussian pulse centered on PHV age:

$$v_g(t) = v_{g,\max} \cdot \exp \left[-\frac{(t - t_{\text{PHV}})^2}{2\sigma_g^2} \right] \quad (5)$$

where $\sigma_g = 1.5$ years for females and $\sigma_g = 1.8$ years for males, reflecting the longer growth spurt duration in males.

3 Results

3.1 The Non-Monotonic K_d - τ^* Relationship

The relationship between derivative gain and critical delay was markedly non-monotonic (Figure 6). For each proportional gain level, there existed an optimal K_d that maximized the critical delay τ^* —that is, an optimal derivative gain that conferred maximal tolerance to sensorimotor delay.

At $K_p = 120$ N·m/rad (the default), the optimal derivative gain was approximately $K_d^* \approx 13$ N·m·s/rad, yielding $\tau^* \approx 96$ ms. Derivative gains below this optimum reduced stability because insufficient rate damping permitted oscillatory buildup. Importantly, derivative gains *above* the optimum also reduced τ^* , because excessive derivative ac-

tion amplified high-frequency noise in the delayed feedback signal, creating a paradoxical destabilization at high K_d .

This non-monotonic relationship defines a “stability corridor” in K_d – τ parameter space: a bounded region within which the postural controller maintains stability. The corridor narrows with increasing delay and widens with increasing K_p .

At higher proportional gains, the optimal derivative gain shifted upward and the maximum tolerable delay increased: at $K_p = 200$, optimal $K_d \approx 20$ and $\tau^* \approx 140$ ms. This suggests that individuals with greater intrinsic trunk stiffness possess a larger stability margin during growth.

3.2 Growth-Phase Vulnerability Window

When the derivative gain was degraded according to Equation 3 using published growth velocity data, the system trajectory in K_d – τ parameter space moved toward the instability boundary during the growth spurt and retreated as growth decelerated (Figure 6).

The vulnerability window—defined as the age interval during which the operating point lay within the unstable region—depended critically on the co-occurrence of two factors: (1) sufficiently rapid growth velocity ($v_g > 6$ cm/yr), producing both K_d degradation and τ elevation, and (2) a baseline delay near the upper end of the normal range ($\tau_0 > 180$ ms). Neither factor alone was sufficient to drive instability: moderate growth with short delay, or slow growth with long delay, both remained within the stable corridor.

Growth-phase transient simulations (Sweep C) confirmed that the system response was highly sensitive to K_d in the vicinity of the stability boundary. At $K_d = 5$ N·m·s/rad (severely degraded), oscillation amplitude exceeded 1 rad, representing complete loss of postural control. At $K_d = 8$ (moderately degraded), transient oscillations of approximately 0.2 rad ($\approx 11^\circ$) were observed, consistent with the magnitude of clinically significant scoliotic curves. At $K_d = 16$ (near-optimal), the system remained stable throughout the simulated growth spurt.

3.3 Sex-Specific Onset Timing

The model predicted distinct vulnerability windows for males and females, arising naturally from the different timing and characteristics of their growth spurts (Figure 6).

For females (PHV at 11.5 years, $v_{g,\max} = 8.3 \text{ cm/yr}$), the effective K_d reached its nadir at approximately 11.5 years, and the combined effect of K_d degradation and τ elevation placed the system trajectory within the unstable region from approximately 10.6 to 11.8 years of age.

For males (PHV at 13.5 years, $v_{g,\max} = 9.5 \text{ cm/yr}$), the corresponding vulnerability window extended from approximately 12.2 to 14.2 years. Although males experience a higher absolute peak growth velocity, their vulnerability window is shifted later by 2 years, consistent with the later median age at AIS detection in boys.

The model also suggested a basis for female preponderance. Given that the female growth spurt is shorter in duration ($\sigma_g = 1.5$ vs. 1.8 years) but occurs at a younger developmental age when neuromuscular adaptation may be less mature, the K_d degradation was modeled as proportionally larger in females. When $\beta_{\text{female}} = 0.65$ and $\beta_{\text{male}} = 0.50$ (reflecting faster adaptive recalibration in older adolescents), the probability of trajectory excursion into the unstable region was 2.8-fold higher in the female model, broadly consistent with the reported 3:1 female-to-male ratio for curves exceeding 30 degrees [1].

3.4 Spontaneous Stabilization at Skeletal Maturity

As growth velocity declined toward zero following PHV, both K_d (Eq. 3) and τ (Eq. 2) returned to their baseline values. The system trajectory retreated from the unstable region, and the postural controller regained adequate stability margin. This provides a mechanistic explanation for the clinical observation that curve progression in AIS typically ceases at skeletal maturity [24], even without treatment. In the model, stabilization occurs not because the curve is “corrected” but because the controller regains the capacity to prevent further progression.

3.5 Brace Mechanism Reinterpretation

Conventional understanding holds that braces work by applying corrective forces to the trunk. Our model suggests an alternative or complementary mechanism: the brace may function as a haptic augmentation of the derivative gain (Figure 6).

A rigid brace in contact with the trunk provides direct cutaneous feedback proportional to the rate of trunk deviation—that is, a mechanical analog of derivative feedback. In the model, adding an external derivative gain $K_{d,\text{ext}} = 3 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$ (representing the brace’s haptic contribution) restored the total effective K_d from 3.2 to 6.2 $\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$ during peak vulnerability, moving the operating point back within the stability corridor.

This reinterpretation is consistent with the clinical observation that brace efficacy depends on compliance (hours of wear per day) rather than corrective force magnitude [25, 26], and that nighttime braces—which provide haptic feedback without substantial corrective force—demonstrate clinical efficacy [27].

3.6 Stochastic Sensitivity

The stochastic analysis (Sweep E) demonstrated that sensory noise substantially modulated the effective stability boundary. At the optimal K_d , the system tolerated noise levels up to $\sigma = 5 \text{ N}\cdot\text{m}$ without destabilization. However, at suboptimal K_d values (near the stability boundary), even modest noise ($\sigma = 1 \text{ N}\cdot\text{m}$) triggered instability in 30% of simulation trials, and high noise ($\sigma = 5 \text{ N}\cdot\text{m}$) triggered instability in over 70% of trials.

This suggests that individuals with noisier proprioceptive signals—potentially due to genetic variants affecting mechanosensory channel function [8, 9]—may have a lower effective stability threshold and therefore greater susceptibility to AIS onset during the growth spurt.

4 Discussion

4.1 Principal Findings

This computational modeling study introduces the concept of a “derivative gain gap”—a transient deficit in the rate-sensing component of trunk postural control that arises during rapid adolescent growth. The model demonstrates that this gap creates a defined instability window through a Hopf bifurcation mechanism, and that the timing and sex-specificity of this window are consistent with the known clinical epidemiology of AIS.

The central insight is that the relationship between derivative gain and delay tolerance is non-monotonic. This means that the postural controller operates within a bounded stability corridor, and that growth-induced changes can push the system outside this corridor even when the proportional gain remains adequate. The “trap” lies in the derivative channel specifically: the rate-sensing pathway is more vulnerable to growth-related disruption than the position-sensing pathway because it requires finer temporal calibration.

Importantly, the supplementary analyses demonstrate that these conclusions are not artifacts of specific parameter choices. The parameter sensitivity analysis (Figure S1) showed that the qualitative predictions held across wide ranges of β and α , and the Monte Carlo analysis (Figure S2) confirmed robustness to individual growth variability. The introduction of adaptive gain recovery (Figure S3) provided a natural explanation for why most adolescents do *not* develop AIS: rapid neuromuscular recalibration closes the derivative gain gap before sustained oscillation can develop. Finally, the two-degree-of-freedom extension (Figure S4) confirmed that the instability mechanism persists in a coupled sagittal-lateral model, supporting the generalizability of the single-plane findings to the three-dimensional clinical deformity.

4.2 A Two-Layer Mechanistic Account

The single-degree-of-freedom model addresses onset: it identifies when postural control becomes unstable and predicts who is at risk. The multi-segment model addresses morphology: it shows where the deformity localizes and why it acquires a three-dimensional

character. Together, they offer a parsimonious account in which a delayed-feedback instability determines onset, regional compliant-span mechanics determines why the deformity localizes to the thoracic and thoracolumbar spans, and vertebral bend-twist coupling determines why lateral deviation becomes axial rotation rather than a simple planar lean.

Stated precisely: *a delayed-feedback instability may determine onset, while regional compliant-span mechanics and bend-twist coupling determine where the deformity localizes and why it becomes three-dimensional.*

Both layers are required; neither is sufficient alone. The pendulum model cannot predict localization at T7–L1 rather than any other spinal level. The multi-segment model, absent the control-theoretic instability, provides no account of timing or sex specificity. The two models are complementary, not redundant.

4.3 Comparison with Existing AIS Theories

The derivative gain gap hypothesis is not in conflict with existing AIS theories but rather provides a mechanistic link between several previously disconnected observations:

- **Biomechanical theories** [4, 5]: The Hueter-Volkmann effect (asymmetric loading driving asymmetric growth) may serve as the *progression* mechanism once the initial deviation is established by the control-theoretic instability proposed here.
- **Proprioceptive theories** [6, 7]: The documented proprioceptive deficits in AIS patients are consistent with the degraded K_d predicted by our model, and may represent the clinical correlate of the derivative gain gap.
- **Growth-related theories** [11, 13]: The correlation between PHV and AIS onset is a direct prediction of our model, not an incidental association.
- **Genetic predisposition**: In our framework, genetic variants (e.g., in *PIEZO2* or other mechanosensory genes) may manifest as altered baseline K_d or increased proprioceptive noise (σ), lowering the threshold for growth-induced instability.

4.4 Testable Predictions

The model generates eight specific, falsifiable predictions:

1. **Proprioceptive velocity sensing:** AIS patients should demonstrate degraded trunk velocity perception (derivative sensing) more consistently than position perception (proportional sensing), testable with psychophysical threshold protocols.
2. **PHV correlation:** The onset of the derivative gain gap should temporally precede curve detection by 6–12 months, testable with prospective longitudinal monitoring of proprioceptive function and growth velocity.
3. **Delay prolongation:** Somatosensory evoked potential latencies should be longer in AIS patients at diagnosis compared with age- and height-matched controls.
4. **Noise amplification:** Postural sway spectral analysis should reveal elevated high-frequency content in pre-scoliotic children during PHV, reflecting inadequate derivative damping.
5. **Brace haptic effect:** Thin, compliant braces providing primarily tactile feedback (with minimal corrective force) should demonstrate measurable improvement in postural stability metrics in AIS patients.
6. **Sex-specific timing:** The proprioceptive deficit should appear approximately 2 years earlier in girls than boys, coinciding with their respective PHV timing.
7. **Maturity-dependent resolution:** Proprioceptive velocity sensing deficits should normalize with skeletal maturity, even in untreated patients with established curves.
8. **Stochastic vulnerability:** Patients with known PIEZO2 polymorphisms should show a higher incidence of AIS and earlier onset age, consistent with the increased noise sensitivity demonstrated in our stochastic simulations.

4.5 Clinical Implications

If validated, this model has several implications for clinical practice:

Screening. Proprioceptive velocity sensing tests administered during annual school screening could identify children entering the derivative gain gap before curves become clinically apparent. This could enable targeted monitoring during the brief high-risk window.

Brace design. If braces function partly through haptic augmentation of K_d , then brace design could be optimized for cutaneous sensory feedback rather than purely corrective force. This might permit lighter, more compliant braces with improved patient compliance.

Physiotherapy. Targeted proprioceptive training focused on velocity sensing (e.g., perturbation-based training) during the peripubertal period could be investigated as a preventive intervention.

4.6 Future Directions

Several extensions of this work are warranted. First, a fully three-dimensional rigid-body model with explicit intervertebral disc and facet joint mechanics would improve biomechanical fidelity beyond the current multi-segment extension; in particular, modelling axial rotation as a driven degree of freedom rather than a coupled consequence would test whether the bend-twist mechanism is sufficient to generate the observed rotational magnitudes without additional torsional loading. Second, incorporation of muscle-specific activation dynamics and co-contraction strategies would improve physiological fidelity. Third, formal sensitivity analysis using Latin hypercube sampling across all parameters would more rigorously characterize model robustness. Fourth, validation against clinical data—specifically, prospective measurement of proprioceptive function, growth velocity, and spinal alignment in a cohort of pre-pubertal children followed through skeletal maturity—would provide the essential empirical test of the hypothesis.

5 Limitations

This study has several limitations that must be acknowledged, along with the steps taken to mitigate their impact.

First, this is a purely computational modeling study. No patient data were collected or analyzed, and the model has not been validated against clinical measurements. To partially address this, we performed a systematic comparison of model predictions against eight independently published clinical observations (Supplementary Table S1), including AIS onset timing, sex ratio, PHV correlation, brace efficacy, proprioceptive deficits, and height association. The model’s predictions were consistent with all eight observations, supporting its plausibility. Nevertheless, prospective clinical validation remains essential.

Second, the model is restricted to a single sagittal plane, whereas clinical AIS is a three-dimensional deformity. To assess whether this simplification invalidates our conclusions, we extended the model to a two-degree-of-freedom coupled sagittal–lateral system (Supplementary Figure S4). The Hopf bifurcation occurred at similar parameter values in the coupled model, and the lateral mode inherited instability from the sagittal mode via mechanical coupling, indicating that the single-plane model captures the essential instability mechanism. A full multi-segment three-dimensional model would add realism but is unlikely to alter the fundamental growth-delay interaction reported here.

Third, the growth velocity curves were modeled as simplified Gaussian pulses based on population-level CDC/WHO reference data. To assess the impact of individual growth variability, we performed a Monte Carlo analysis with 1,000 simulated individuals drawn from published distributions of PHV age (± 1 year), PHV magnitude (± 1.5 cm/yr), and baseline delay (± 20 ms) (Supplementary Figure S2). The qualitative predictions—female-before-male onset, PHV-correlated vulnerability, spontaneous stabilization—were robust across the sampled population, and the fraction of individuals experiencing instability was consistent with published AIS prevalence estimates when adaptation rate variability was included.

Fourth, the degradation function for K_d (Eq. 3) was postulated based on plausibility rather than directly measured. A full parameter sensitivity analysis (Supplementary

Figure S1) demonstrated that the qualitative predictions of the model were preserved across a wide range of β (0.3–0.8) and α (5–20 ms per cm/yr), indicating that the conclusions do not depend on precise parameter tuning. Nonetheless, direct measurement of proprioceptive gain during adolescent growth remains a priority for clinical validation.

Fifth, the baseline model does not account for neuromuscular adaptation, which would allow some individuals to recalibrate their control gains and avoid the instability window. To address this, we introduced an adaptive gain recovery term representing CNS recalibration at variable rates (Supplementary Figure S3). The extended model showed that fast adapters (recalibration rate $> 0.3/\text{year}$) avoided instability entirely, while slow adapters (rate $< 0.1/\text{year}$) experienced prolonged vulnerability windows. This naturally explains both the low overall prevalence of AIS (2–4%) and the observation that not all rapidly growing adolescents develop scoliosis: individual variation in adaptation rate acts as a gating mechanism.

Sixth, the inverted pendulum model uses a rigid-body approximation with a single lumbar pivot. The spine is a multi-segmental, viscoelastic structure, and this simplification reduces biomechanical fidelity. However, the delayed-feedback instability we describe is a property of the control loop architecture, not the plant dynamics per se, and would be expected to persist in more detailed musculoskeletal models with similar delay and gain characteristics.

Seventh, the differential β values used for males and females were assumed rather than derived from independent evidence. Published data support an indirect basis for this assumption: female adolescents show earlier proprioceptive maturation plateaus [19], different muscle activation timing patterns, and a narrower growth spurt window [23], all of which are consistent with a higher effective β in females. However, direct longitudinal measurement of sex-differential proprioceptive gain degradation during growth has not been performed and remains a key prediction of this model.

6 Conclusions

We present a control-theoretic model of trunk postural control in which growth-velocity-dependent degradation of the derivative gain creates a transient “derivative gain gap” during adolescence. This computational model reproduces several hallmark features of AIS epidemiology: adolescent onset, correlation with peak height velocity, sex-specific timing, female preponderance, and spontaneous stabilization at skeletal maturity. The model further suggests a mechanistic reinterpretation of brace therapy as haptic augmentation of the degraded rate-sensing channel. These findings are consistent with the hypothesis that AIS onset reflects a transient control-theoretic instability rather than a static structural defect. Clinical validation through prospective proprioceptive testing during the peripubertal growth spurt is warranted to test the specific predictions generated by this model.

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Conflict of Interest

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Figure Legends

Figure 1. Two-dimensional stability map in K_d - τ parameter space. Heat map showing the amplitude ratio (a measure of instability) as a function of derivative gain (K_d , y -axis) and sensorimotor delay (τ , x -axis) at $K_p = 120 \text{ N}\cdot\text{m}/\text{rad}$. The Hopf bifurcation boundary (white contour) separates stable (blue) from unstable (red) regions. Note the non-monotonic shape: both excessively low and excessively high K_d values reduce the tolerable delay. The optimal $K_d \approx 13 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$ maximizes τ^* .

Figure 2. Growth-phase trajectory through the stability landscape. The operating point of the postural controller (black curve) is plotted as it evolves during the female adolescent growth spurt. As growth velocity increases toward PHV, K_d decreases (downward movement) while τ increases (rightward movement), driving the trajectory across the Hopf bifurcation boundary into the unstable region (shaded red). After PHV, growth deceleration returns the trajectory to the stable region. The brief excursion into the unstable zone constitutes the “derivative gain gap.”

Figure 3. Sex-specific vulnerability windows. Panel (A): Modeled growth velocity curves for females (solid line, PHV at 11.5 years) and males (dashed line, PHV at 13.5 years). Panel (B): Corresponding effective derivative gain $K_d(t)$ showing the growth-dependent degradation trough. Panel (C): Instability index (distance from the Hopf boundary) as a function of age. Shaded regions indicate periods during which the instability index exceeds zero, corresponding to the vulnerability window: 10.6–11.8 years for females and 12.2–14.2 years for males.

Figure 4. Brace mechanism reinterpretation: haptic K_d augmentation. Panel (A): Without brace—the degraded K_d at PHV places the operating point outside the stability corridor (red dot). Panel (B): With brace—the external haptic feedback contributes an additive $K_{d,\text{ext}}$, restoring the total effective K_d and moving the operating point back inside the stability corridor (green dot). Panel (C): Time-domain simulation showing oscillation amplitude with (green) and without (red) brace augmentation during the modeled growth spurt.

Figure 5. Stochastic sensitivity of the derivative gain gap. Probability of instability (defined as $\max|\theta| > 0.5$ rad across 30 simulation trials) as a function of K_d at three noise levels ($\sigma = 0, 1, 5$ N·m) with $\tau = 60$ ms. At optimal $K_d \approx 13$, the system resists even high noise. At suboptimal K_d values near the stability boundary, noise sharply increases the probability of instability, suggesting that individuals with noisier proprioceptive channels (e.g., PIEZO2 variants) face elevated AIS risk.